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The Editors
Digital Discovery
Royal Society of Chemistry

Dear Editors,
16 September 2026

We submit the manuscript entitled “**Deconstructing Out-of-Distribution Extrapolation in Molecular Deep Learning: Structural Complexity, Topological Rescue, and Conformal Triage on African Antimalarial Natural Product Scaffolds**” for consideration as a Full Paper in *Digital Discovery*.

This study speaks directly to your journal’s data-first mission. Rather than another passive benchmark reporting fingerprint dominance, we deliver a mechanistic explanation for graph neural network extrapolation failures and an actionable decision framework for virtual screening under distribution shift. On a frozen panel of 19 836 African antimalarial natural products and synthetic derivatives, evaluated across 25 fold–seed replicates per model, we establish three contributions:

First, we localise the ECFP4–RF advantage strictly to the structural extrapolation tail (nearest-neighbour similarity < 0.40) and identify its mechanism: the 1-Weisfeiler–Lehman expressivity ceiling of message-passing networks on high- Fsp^3 polycyclic scaffolds. Fusing persistent-homology topological features (GIN–TFP) recovers up to 36 % of that gap.

Second, we operationalise these dynamics as a Distance-Aware Conformal Triage Filter that routes candidates to the right model by distance regime, improving prospective screening precision by **+14.2 %** on a disjoint 22 267-molecule ChEMBL transfer panel.

Third, we show that fold-independent weight restoration in molecular transformers is mandatory to prevent cross-fold leakage — a reproducibility standard for the community.

Data and code: all splits, predictions, and analysis scripts are archived on Zenodo (DOI 10.5281/zenodo.19608875, record reserved pending upload completion) and GitHub (https://github.com/taamangtchu/Project5_GNN_Transformer_DrugDiscovery). Editors and referees will be given full access for peer review, and the DOI will be published before acceptance, per your data policy.

The work is original, is not under consideration elsewhere, has been approved by all authors, and there are no conflicts to declare.

Suggested reviewers:

1. **Prof. Kelin Xia** (Nanyang Technological University, Singapore) — Topological deep learning, persistent homology, molecular graph representations. Email: xiakelin@ntu.edu.sg
2. **Prof. Fidele Ntie-Kang** (University of Buea, Cameroon / Martin Luther University Halle-Wittenberg, Germany) — African Natural Products Database (ANPDB), natural product cheminformatics. Email: fidele.ntie-kang@uni-halle.de
3. **Prof. Carsten Eickhoff** (University of Tübingen, Germany) — Molecular representation benchmarking, out-of-distribution generalisation. Email: c.eickhoff@uni-tuebingen.de

We thank you for your consideration.

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